Prediction of Bilingual Proficiency through Eye-Tracking Analysis of Mandarin Chinese-English Code-Switching: A Machine Learning Approach

Selina Hui, Lily Wu, Asha Dukkipati, and Sienna Jackson

Supplemental Information & Figures

1. Background Info & Literature Review

Key Term Definitions:

- **Code-Switching**: Alternating between two languages (e.g., Chinese and English) within a single sentence or conversation.
- **Late-Stage Eye-Movement Metrics**: Eye behaviors that occur after the initial reading of a word, often involving re-reading, regressions, and integration processes.
- **Early-Stage Eye-Movement Metrics:** Measures that capture immediate, initial processing of text during first-pass reading (e.g., first fixation duration, first-pass gaze duration), reflecting rapid lexical access and initial word recognition.
- **Saccade**: A rapid eye movement between two points during reading.
- **Fixation:** A brief pause of the eyes on a specific word or region, allowing for visual information to be processed. Fixations typically last between 200–300 milliseconds during reading.
- **Skipping Behavior**: The act of moving past a word without fixating on it, typically reflecting fluent or predictive reading.
- **Cognitive Effort** (in Reading): The mental processing load involved in decoding and understanding text, often reflected by longer fixations and more regressions.
- **Regression** (Eye-Tracking): A backward eye movement to re-examine earlier parts of the text due to difficulty or confusion.
- **Random Forest Classifier**: A machine learning model that makes predictions by aggregating decisions from many individual decision trees to improve accuracy and robustness.
- **Overfitting**: A modeling issue where the model fits the training data too closely, including noise, and performs poorly on new data.
- **Cross-Validation**: A method for testing a model's generalizability by repeatedly training and testing on different data subsets.
- **Area Under the Curve (AUC)**: A performance metric for classifiers; higher AUC values indicate better ability to distinguish between classes (e.g., high vs. low proficiency).

Eye Movement Feature Definitions:

Eye-Movement Measures		Abbreviations	Definitions
Temporal	First fixation duration	FFD	Duration of the first fixation event that was within the current interest area.
	Second fixation duration	SFD	Duration of the second fixation in the current interest area, regardless of run.
	Gaze duration	GD	Dwell time of the first run (i.e., the sum of the duration of all fixations in the first run of fixations within the current interest area).
	Regression path duration	RPD	The amount of time beginning with the first fixation on the critical stimulus until the eyes cross the right-hand boundary of the region
	Total dwell time	TDT	The sum of the duration across all fixations that fell in the current interest area.
Spatial	Saccade amplitude	SA	Amplitude (in degrees of visual angle) of the first saccade entering into the current interest area.
	Fixation count	FC	Total number of fixations falling in the interest area.
	Skip	PS1	An interest area is considered skipped (i.e., IA_SKIP = 1) if no fixation occurred in first-pass reading.
	First run fixation count	FFC	Number of all fixations in a trial falling in the first run of the current interest area.
	Regression in count	RI	Number of times interest area was entered from later parts of the sentence.
	Regression out count	RO	Number of times interest area was exited to the earlier parts of the sentence before leaving the current interest area in a forward direction.

Table 1: Definitions and Abbreviations of the Nine Eye-Movement Measures (Wang *et al.,* 2025)

Hypothesis:

Bilinguals with lower L2 proficiency are expected to exhibit increased late-stage eye-movement activity, characterized by more fixations, longer regression paths, and extended reading times, as they consciously re-evaluate text during reading. In contrast, bilinguals with higher L2 proficiency are anticipated to show more efficient eye-movement patterns, with fewer fixations, shorter regression paths, and quicker progression through the text. For saccadic behavior, higher-proficiency readers are expected to make longer forward saccades and skip more text, while lower-proficiency readers will exhibit shorter, word-by-word saccades reflecting slower and more effortful lexical processing.

2. GitHub Code Availability

Full code for preprocessing, feature engineering, and Random Forests training is available here.

3. Initial Visualizations

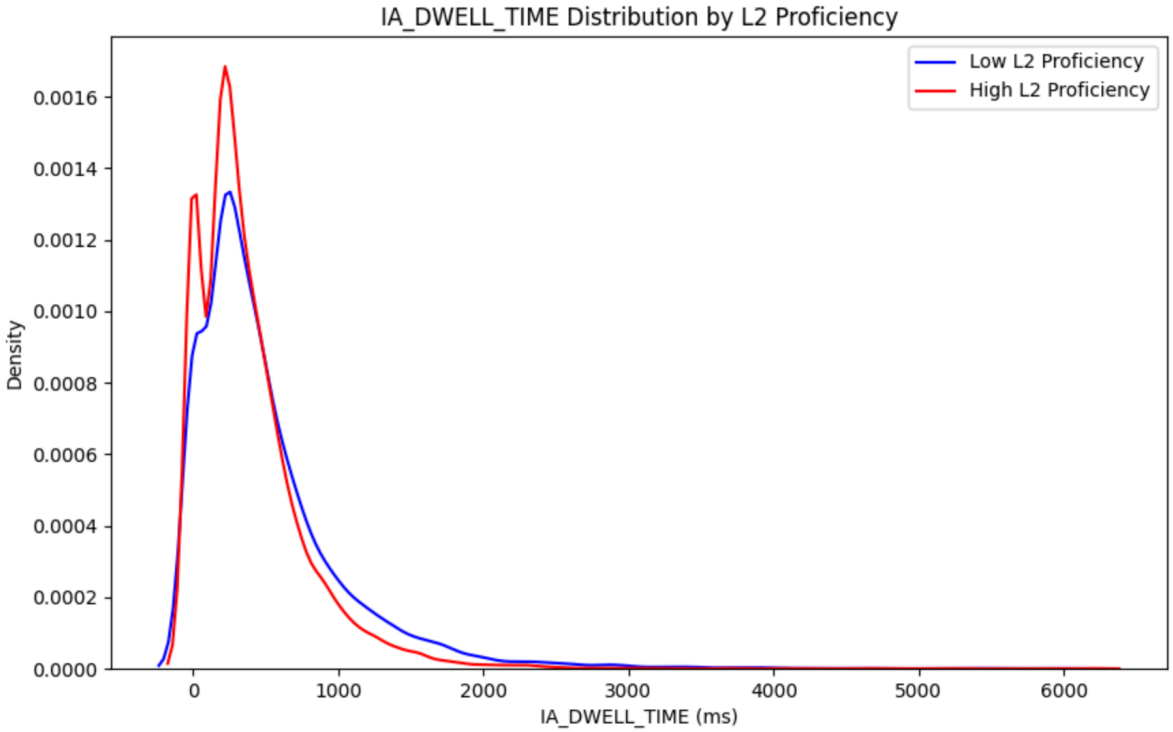


Figure S1: Distribution of Dwell Time in High vs. Low L2 Proficiency participants

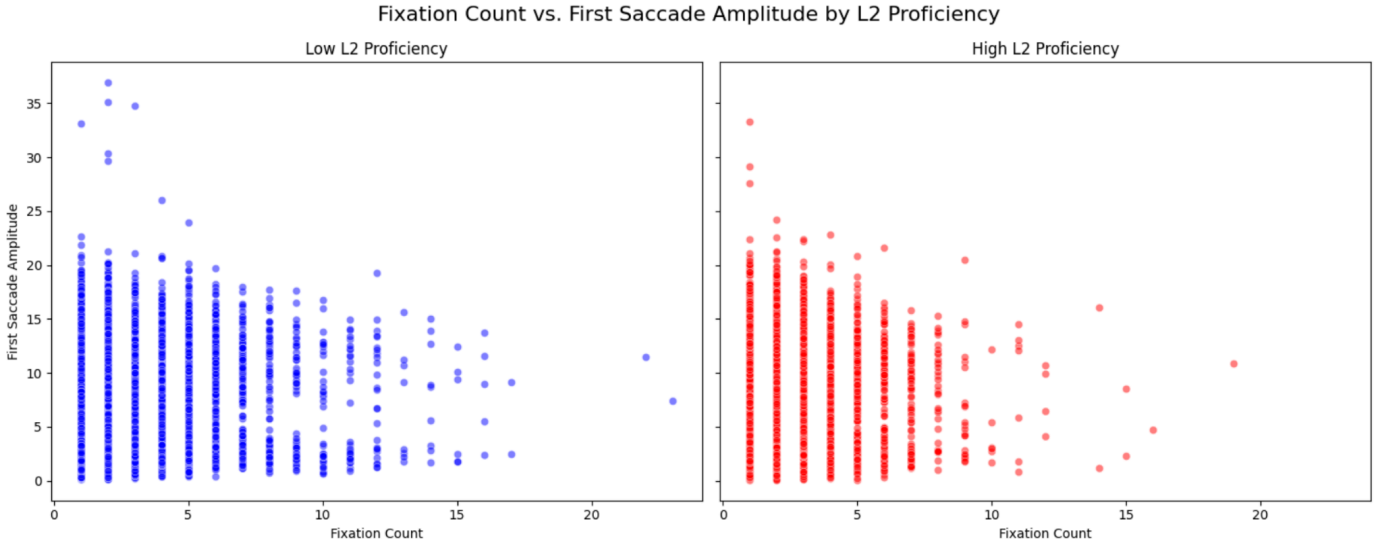


Figure S2: Comparison of Fixation Count vs. First Saccade Amplitude across L2 Proficiency groups

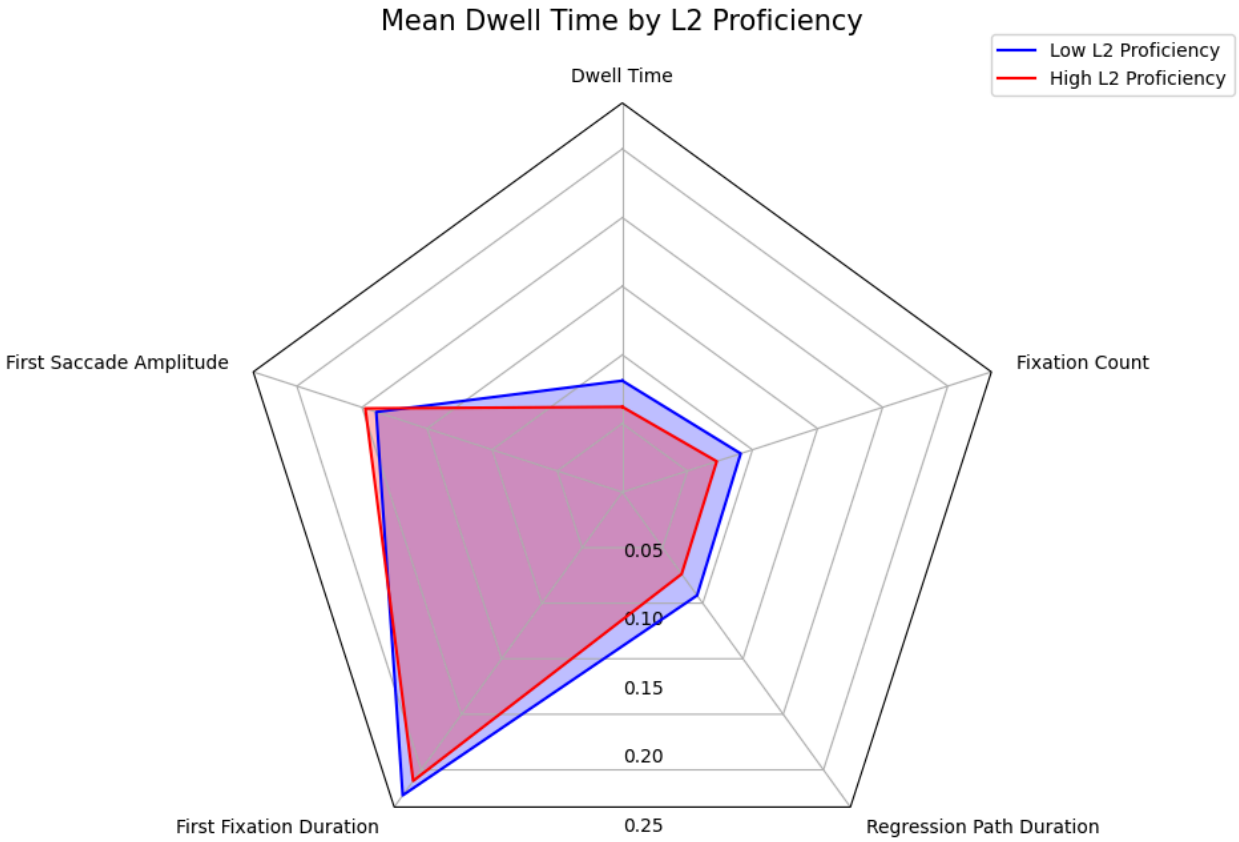


Figure S3: Mean Dwell Time Values for Low vs. High L2 Proficiency. Participants with lower L2 proficiency spent more time per fixation on average, consistent with less fluent reading behavior.

4. Data Preprocessing

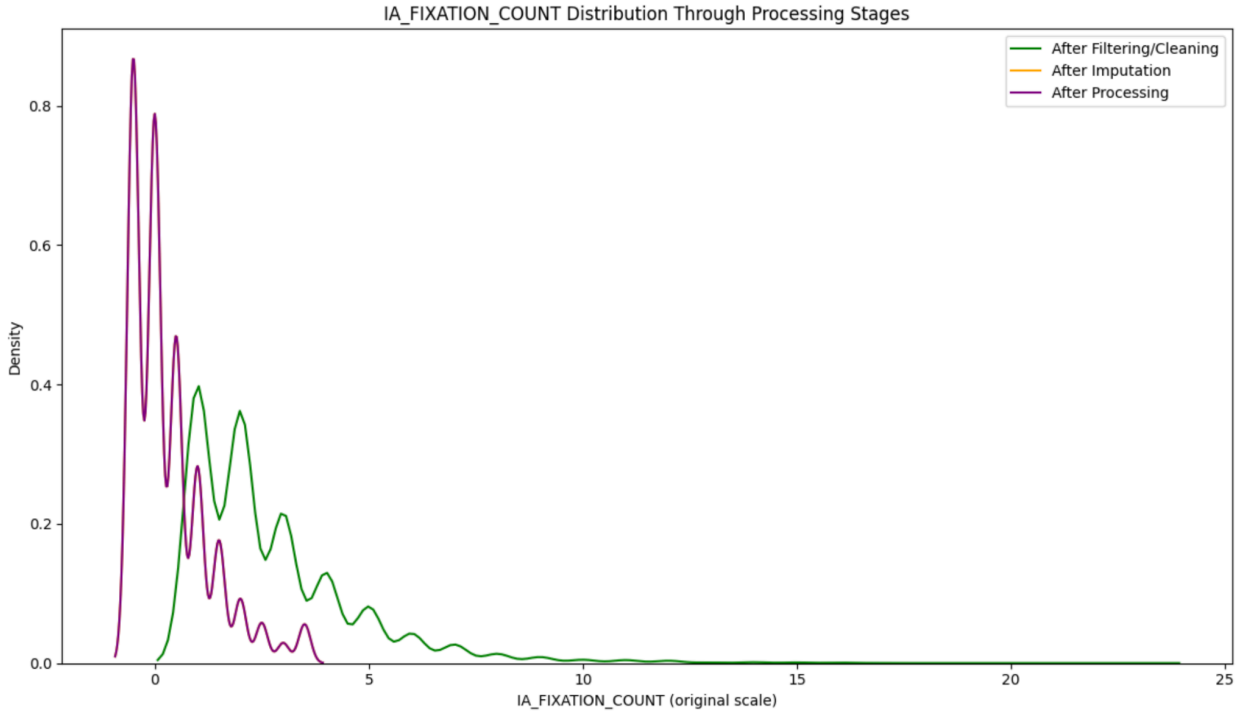


Figure S4: Distribution of Fixation Count After Preprocessing Stages.

Note: The distribution after imputation closely matches the post-processing distribution and is overlaid by the purple line.

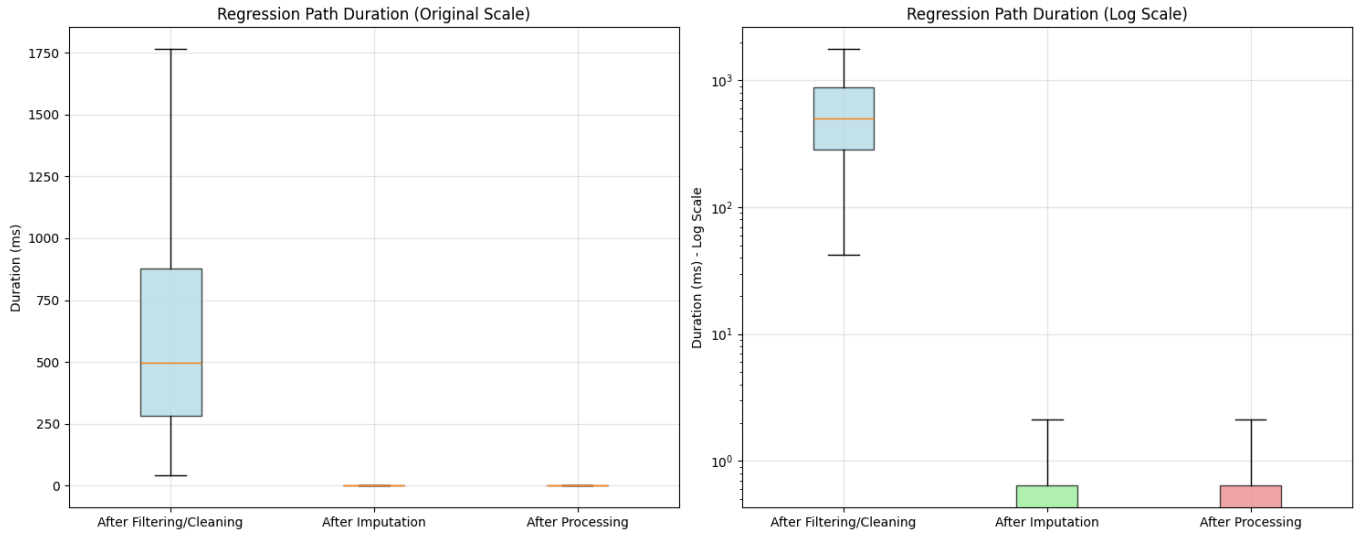


Figure S5: Distribution of Regression Path Duration in Original & Log Scale After Preprocessing Stages

```
=== Preprocessing Metrics ===

Null Values:
Initial: 43615 (9.0%)
Final: 3483 (1.6%)
Reduction: 92.0%

Class Distribution:
Before balancing: L2 PROFICIENCY
L    0.5
H    0.5
Name: proportion, dtype: float64
After balancing: L2 PROFICIENCY
1    0.50744
0    0.49256
Name: proportion, dtype: float64

Outlier Handling:
Total outliers across all categories: 6869 (26.8%)

Summary Statistics:

After Filtering/Cleaning:
Mean: 2.72
Std: 2.01
Min: 1.00
Max: 23.00
Count: 11290

After Imputation:
Mean: 0.34
Std: 0.92
Min: -0.50
Max: 3.50
Count: 11290

After Processing:
Mean: 0.34
Std: 0.92
Min: -0.50
Max: 3.50
Count: 11290
```

Figure S6: Summary Metrics After Preprocessing Stages

5. Feature Engineering & Importance Analysis

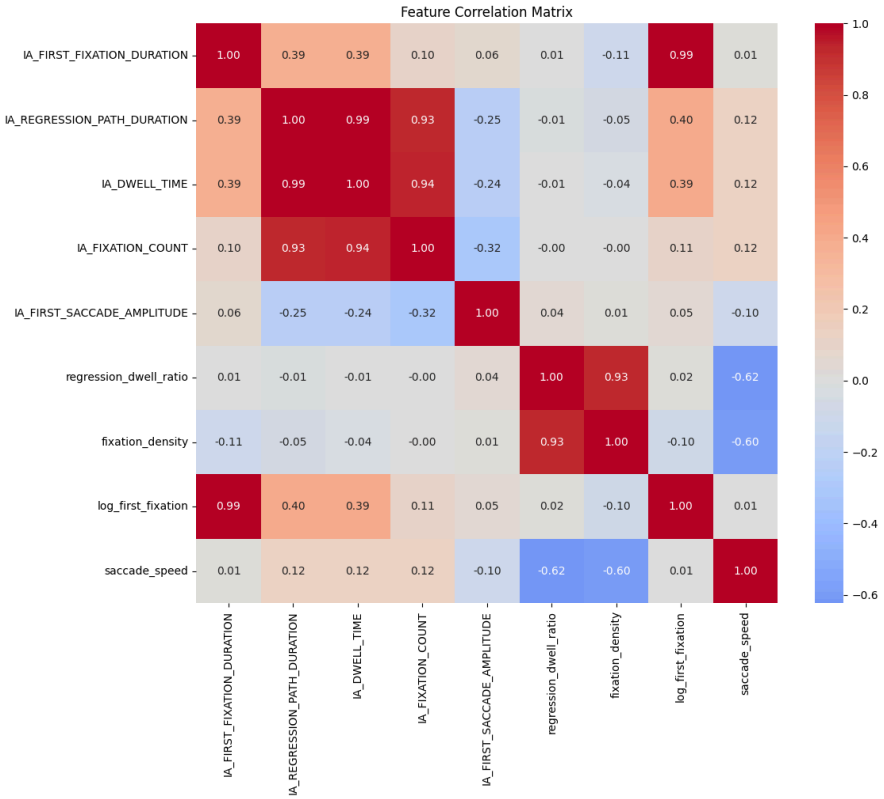


Figure S7: Correlation Matrix of Available Features to Predict English L2 Proficiency. Late-stage metrics (dwell time, regression path duration) showed strongest positive correlations with lower L2 proficiency.

```
=== Feature Analysis ===

All Available Features:
1. IA_REGRESSION_PATH_DURATION: 0.163 SELECTED
2. IA_FIXATION_COUNT: 0.162 SELECTED
3. IA_DWELL_TIME: 0.157 SELECTED
4. fixation_density: 0.111 SELECTED
5. regression_dwell_ratio: 0.094 SELECTED
6. log_first_fixation: 0.082
7. IA_FIRST_FIXATION_DURATION: 0.082
8. IA_FIRST_SACCADE_AMPLITUDE: 0.075
9. saccade_speed: 0.074
10. CONDITION: 0.000

Selected Features for Training:
1. IA_DWELL_TIME
   Importance: 0.157
   Correlation with target: 0.499
   Description: Total time spent looking at a word
2. IA_FIXATION_COUNT
   Importance: 0.162
   Correlation with target: 0.460
   Description: Number of fixations on a word
3. IA_REGRESSION_PATH_DURATION
   Importance: 0.163
   Correlation with target: 0.495
   Description: Time spent in regression movements
4. fixation_density
   Importance: 0.111
   Correlation with target: -0.098
   Description: Number of fixations per unit time
5. regression_dwell_ratio
   Importance: 0.094
   Correlation with target: -0.098
   Description: Ratio of regression time to total dwell time
```

Figure S8: Summary Table of Features, Importance, and Correlation. Dwell Time and Regression Path Duration were the most influential predictors in the Random Forest classifier, consistent with increased cognitive load in low-proficiency readers.

6. Model Training & Evaluation

```
Fitting 15 folds for each of 50 candidates, totalling 750 fits

=== Best Overfitting-Reducing Configuration ===
Key parameters for generalization:
Best max_depth: 5
Best min_samples_split: 10
Best min_samples_leaf: 8
Best max_features: 0.5

Best F1 Score (CV): 0.70
Final test score: 0.84
Accuracy: 0.69
              precision    recall  f1-score   support

           0       0.67      0.75      0.71        16
           1       0.71      0.62      0.67        16

    accuracy                           0.69        32
   macro avg       0.69      0.69      0.69        32
weighted avg       0.69      0.69      0.69        32
```

```
=== Overfitting Analysis ===
Train Accuracy: 0.86 (±0.02)
Test Accuracy:  0.72 (±0.09)

Train F1: 0.86 (±0.02)
Test F1:  0.72 (±0.09)
Train Accuracy: 0.88
Test Accuracy: 0.69
```

Figure S9: Model Results Summary of Best Configuration to Reduce Overfitting & Evaluation Results

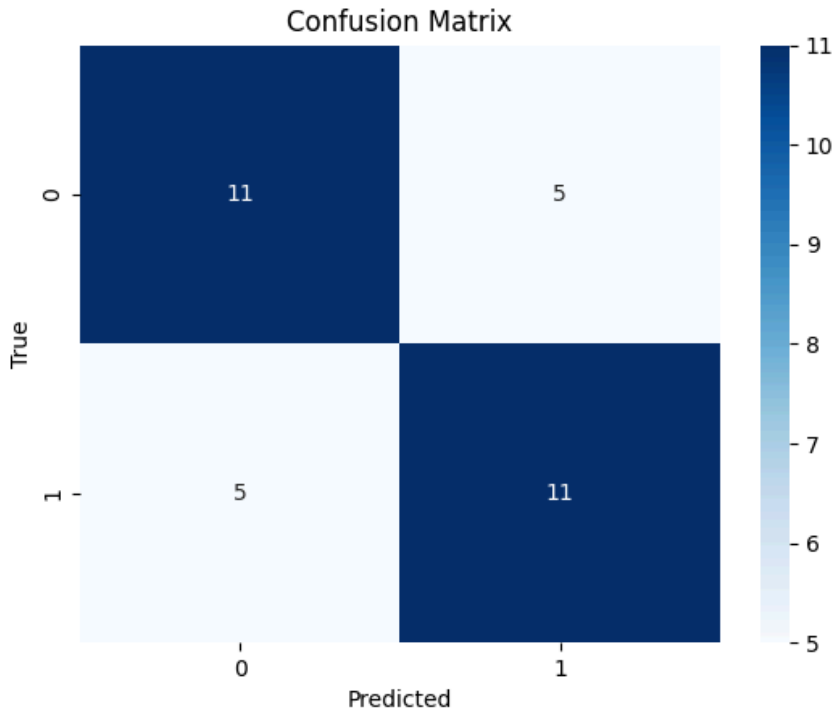


Figure S10: Confusion Matrix Heatmap of True vs. Predicted Labels for High vs. Low L2 Proficiency

7. Other Stages

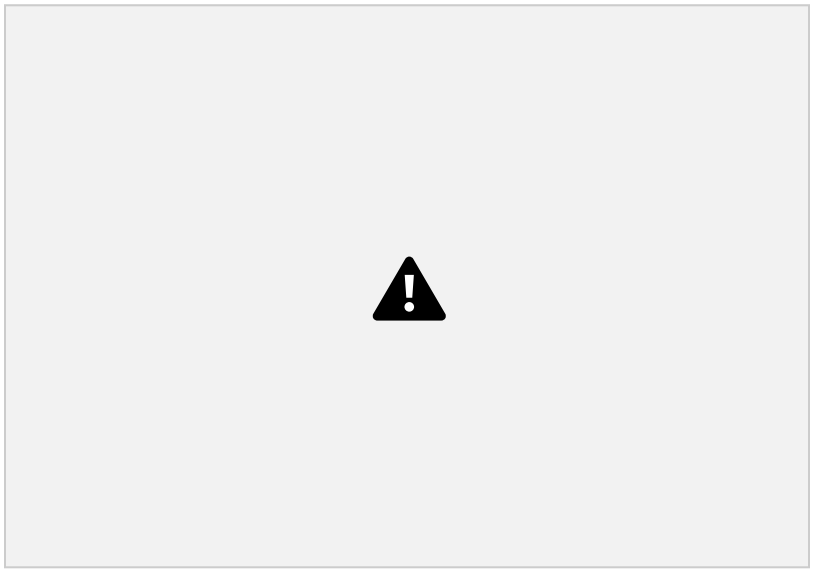


Figure S11: Tree Map #1 Visualization of Random Forests Classification Logic. Displays how eye-movement features contributed to decision splits for classifying L2 proficiency.

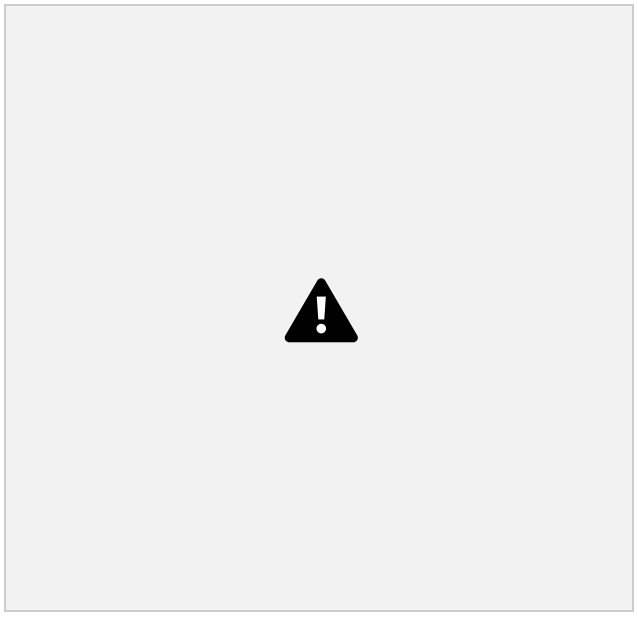


Figure S12: Tree Map #2 Visualization of Random Forests Classification Logic. Displays how eye-movement features contributed to decision splits for classifying L2 proficiency.

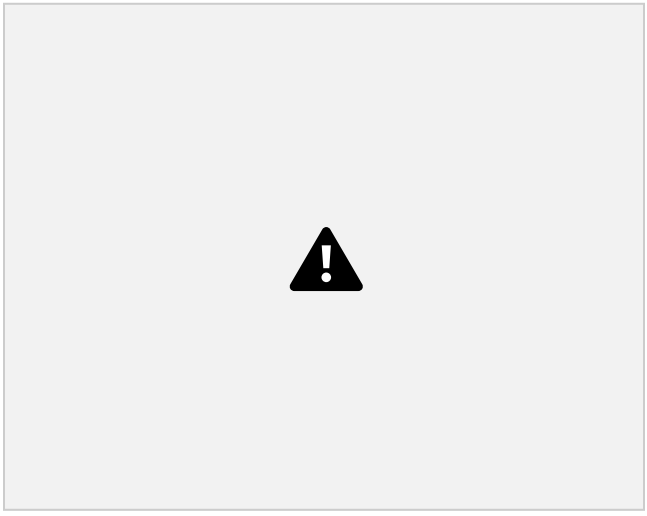


Figure S13: Tree Map #3 Visualization of Random Forests Classification Logic. Displays how eye-movement features contributed to decision splits for classifying L2 proficiency.

Summary of Supplementary Information & Figures:
Overall, supplemental analysis of feature distributions, preprocessing impact, and model evaluation supports the main finding that late-stage eye-movement metrics are predictive of L2 proficiency in bilingual readers. These analyses further validate the model's generalization and the feature importance patterns observed.